

Statistics Practice Set S33

10 questions | No hints | Application-focused | No duplicate questions

Q1 — Conditional Probability: Customer Support

A company receives support requests through three channels:

Channel	Requests	Resolved
Chat	2,000	1,600
Email	1,500	900

Find:

- 1. Total requests
 - 2. Total resolved requests
 - 3. $P(\text{Resolved} \mid \text{Chat})$
 - 4. $P(\text{Chat} \mid \text{Resolved})$
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Q2 — Independent vs Dependent: Email Campaign

A company compares customers who received an email with those who did not.

Group	Purchased	Did Not Purchase
Received Email	240	960
Did Not Receive Email	120	880

Find:

1. Purchase rate among customers who received the email.
 2. Purchase rate among customers who did not receive the email.
 3. Are purchase and receiving the email **independent or dependent**?
 4. Give a one-sentence business interpretation.
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Q3 — Addition Rule: Customer Platforms

Out of **5,000** customers:

- 3,000 used the mobile app.
- 2,200 used the website.
- 800 used **both**.

Find:

1. $P(\text{App})$
 2. $P(\text{Website})$
 3. Number of customers who used **at least one** platform.
 4. Number of customers who used **neither** platform.
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Q4 — Complement: Customer Retention

A company has a monthly churn rate of **11%**.

Find:

1. Non-churn rate.
 2. Expected number of non-churned customers out of **12,000** customers.
 3. Is this number guaranteed to occur exactly? Explain briefly.
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Q5 — Mutually Exclusive vs Overlapping

Consider online transactions:

- Event A = Transaction amount is **below ₹1,000**
- Event B = Transaction amount is **between ₹500 and ₹2,000**
- Event C = Transaction amount is **above ₹3,000**

Answer:

1. Are A and B mutually exclusive?
 2. Are A and C mutually exclusive?
 3. If $A \cap B$ is non-empty, what does that mean in simple words?
 4. What is $A \cap C$?
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Q6 — Bayes' Theorem: Fraud Detection

A payment system processes:

Transaction	Total	Fraudulent
Online	12,000	360
Offline	3,000	90

Find:

1. Total transactions.
 2. Total fraudulent transactions.
 3. $P(\text{Fraud} \mid \text{Online})$
 4. $P(\text{Online} \mid \text{Fraud})$
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Q7 — Rate vs Count: Marketing Campaigns

Three campaigns produced:

Campaign	Customers Reached	Purchases
A	12,000	840
B	5,000	450
C	2,000	220

Find:

1. Conversion rate of A.
 2. Conversion rate of B.
 3. Conversion rate of C.
 4. Which campaign generated the **highest number of purchases**?
 5. Which campaign had the **highest conversion rate**?
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Q8 — Analyst Interpretation

An analyst says:

“Campaign A is definitely the best because it generated the highest number of purchases.”

Do you agree or disagree?

Give **one reason** using **conversion rate**.

Q9 — Reverse Probability: Employee Skills

A company has:

Department	Employees	Python Skilled
Data Analytics	500	300
Finance	700	140
Marketing	800	80

Find:

1. Total employees.
 2. Total Python-skilled employees.
 3. $P(\text{Python} \mid \text{Data Analytics})$
 4. $P(\text{Data Analytics} \mid \text{Python})$
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Q10 — Full Business Case: Churn

A subscription company has:

Plan	Customers	Churned
Basic	6,000	720
Premium	3,000	270
Enterprise	1,000	40

Find:

1. Churn rate for each plan.
2. Which plan has the highest churn rate?
3. Can the company conclude that this plan **causes** higher churn? Explain why or why not.